

BUILD

Discharge notes
328,839 MIMIC-IV

Qwen extraction
cause → effect triples

UMLS normalization
merge synonyms to CUIs

Candidate causal KG
4.3K concepts / 19K candidates

AUDIT

Which validation signals deserve trust?

SemMedDB is an independent literature reference, not causal ground truth.

WORKS

Co-occurrence lift

EHR support signal
filters weakly supported co-mentions

FAILS

Coding-order time

Weak direction signal
54.2% vs 73.6%; coding order is not onset

PARTIAL

Literature evidence

Independent reference
74% direction agreement; partial coverage

Takeaway: lift supports candidates;
coding-order time should not orient edges.

APPLY

EHR-supported
causal-candidate KG

Support-aware causal
RAG + demotion

Held-out patient
causal question

Unsupported hallucinations
17.3% → 6.1%